

Dheeraj C.

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EDUCATION

Birla Institute of Technology, Ranchi - Bachelor's, Computer Engineering 2023 - 2027

ACHIEVEMENTS / CERTIFICATIONS

- Biscuit - open-source AI IDE · 268+ GitHub stars · pip install biscuit-editor.
- LeetCode - 409 solved · peak rating 1893.
- Contributions to Zulip, prometheus-operator, and KubeVirt.
- Hackathon Wins: Monad Blitz Bangalore V4.
- Professional Certifications: Meta Backend Developer, IBM Full Stack, Google Cloud AI Professional.

PROFESSIONAL EXPERIENCE

AI/ML Engineer Intern · Hooman Digital LLP | Live Product July 2025 - Sept 2025

Kolkata, India

- Built an LLM pipeline extracting and tracking financial tickers from live market data with 98% accuracy.
- Replaced WebSockets with Server Sent Events (SSE) for one-way log streaming, reducing infrastructure cost by 40%.
- Designed an event-driven backend using AWS SQS to absorb traffic spikes while serving 400+ monthly active users.
- Engineered scalable backend infrastructure components, improving system throughput and reducing query latency by 15%.
- Collaborated with frontend and product teams to design, implement, and ship production features across multiple releases.

Software Development Engineer Intern · OZI | Live Product Nov 2024 - Jan 2025

Gurgaon, India

- Built core MVP features powering a production platform with 15,000+ curated products and 100K+ Android downloads.
- Developed deep-linking and search APIs to improve content discovery and user navigation across the application.
- Optimized backend data access through query tuning and caching, reducing API response latency by 20%.
- Worked directly with the founding team to rapidly prototype, ship, and iterate on production features in a startup.

RESEARCH EXPERIENCE

A Transformer-Based Approach for Raw EEG Classification May 2025 - July 2025

Research Intern at Dept. of Computer Engineering, NIT Calicut · Calicut, India

- Improved biometric classification accuracy by 25% by building state-of-the-art CNN-Transformer fusion models.
- Engineered 5+ EEG classifier fusion models handling spatial-temporal multi-channel tensors using PyTorch.

PROJECTS / OPEN SOURCE

Biscuit - IDE | Python, LangChain, RAG, MCP, React, Pydantic Live Project

- Built a modular IDE using microkernel architecture, featuring source control, pseudoterminals, and extension marketplace.
- Introduced semantic code intelligence for 20+ languages using Tree-sitter and LSP (268+ GitHub stars; pip install biscuit-editor).
- Optimized global search latency under 100ms for folders exceeding 1M+ lines, using multi-threaded Ripgrep.

Hiroshi OS - AI Gateway | Rust, Tokio, Tailscale, MCP, WebSockets, Docker Live Project

- Engineered a pluggable AI gateway with sandboxing, cron scheduling, and secure HTTP RPC services with authentication.
- Reduced token usage by 70% by implementing an incremental diff pipeline and vector retrieval, eliminating full-file rewrites.

Elo - Real-Time Competition Platform | TypeScript, Bun, React Native, Expo, Protobuf, Docker Live Project

- Engineered a math competition app using React Native, Proto, Bun delivering round-trip latency under 40ms.
- Implemented an anti-cheat engine tracking rapid inputs (less than 120ms) into honeypot sandbox to issue hardware bans.

BILL OS - x86_64 Operating System | Rust, Assembly, Limine, QEMU, GDB, Cargo Live Project

- Built a 64-bit operating system, with bootstrapping, interrupt handling, and a modular kernel spanning 10+ subsystems.
- Engineered a kernel with device drivers, shell, ELF executable loading, validated through 1000+ QEMU boot cycles using GDB.

TECHNICAL SKILLS

Programming Languages: Python, Go, Rust, TypeScript, C++, SQL, Bash

Libraries / Frameworks: Django, FastAPI, Tauri, Electron, React, React Native, gRPC, Protobuf, LangChain, Tokio

Tools / Platform: AWS (SQS, EC2, S3, DynamoDB), Docker, Kafka, Tailscale, WSL, Git, Jujutsu

Databases: PostgreSQL, CockroachDB, Redis, pgvector, Pinecone